



COMSATS University Islamabad, Lahore Campus

Department of Computer Engineering

Quiz # 4

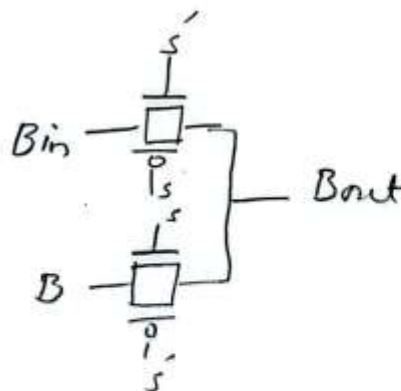
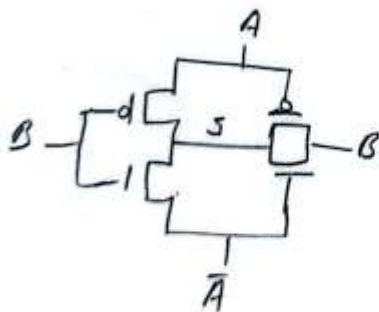
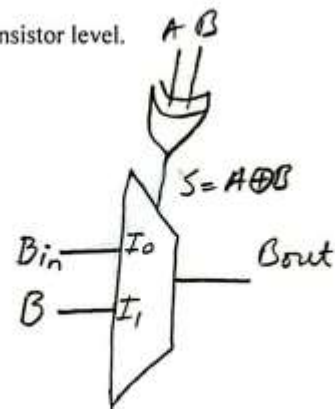
Course Title:	VLSI Design	Course Code:	CPE434	Credit Hours:	4(3,1)
Course Instructor:	Dr. Muhammad Naeem Awais	Programme Name:	BS Computer Engineering		
Semester:		Batch:		Date:	19 th May 2026
Time Allowed:	10 Minutes		Maximum Marks:	5	
Student's Name			Registration. No.:		

Question # 1:

(PLO3-CLO3-C5)

Implement the Borrow output of a Full Subtractor using 2x1 Mux at transistor level.

A	B	Bin	D	Bout	
0	0	0	0	0	$B_{out} = Bin$
0	0	1	1	1	
0	1	0	1	1	$B_{out} = B$
0	1	1	0	1	
1	0	0	1	0	$B_{out} = B$
1	0	1	0	0	
1	1	0	0	0	$B_{out} = Bin$
1	1	1	1	1	



Invertible Inverter Design of
XOR-gate